Milestone-3 Report: Safety Verification of Chiron IR using Constrained Horn Clauses

Project Overview

Program verification provides mathematical guarantees that software behaves correctly, unlike testing, which can find bugs but never prove their absence. While the Chiron framework offers support for dynamic analysis techniques such as fuzzing, symbolic execution, and spectrum-based fault localization, it currently lacks the capability to *prove* that a program satisfies a given specification for *all* possible inputs.

This project bridges the gap between testing and verification in Chiron by implementing a PDR-based safety verification engine. We target properties natural to turtle graphics programs, such as spatial bounds on the turtle's position and invariants over program variables.

Team Information

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AI Use in the Project

In this project, we utilized AI tools to enhance our workflow and improve the quality of our work.

1. **Are you using AI for coding your implementation?** Yes, we used AI tools in this project.
2. **If yes, what models are you using?** We used models like OpenAI's GPT-5.2 Codex and GitHub Copilot.
3. **If using AI, how are you using AI and what percentage (approx) of your code is AI generated?** We used AI primarily for generating boilerplate code, suggesting improvements, and debugging. Approximately $40\% - 50\%$ of our code was generated or assisted by AI tools.
   - AI was primarily used for generating boilerplate code and usage of correct syntax, which was then reviewed and modified by us to fit our specific requirements.
   - We also used AI for testing, where it helped in generating test cases and suggesting improvements to our existing tests.
   - Additionally, AI was used for the generation of documentation and figures in the report, where it assisted in summarizing the directory structure and usage guide, as well as making the figures in the report with careful and structured instructions about the layout.

Declaration

- We declare that the work presented in this report is our own and has been carried out in accordance with the guidelines set out by the course.
- We have not copied any part of this report from any other source, and we have not submitted this report to any other course or institution for assessment.
- We understand that any violation of this declaration may result in disciplinary action.